VEDANT GANDHI
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EDUCATION

Carleton University - BCS (Honours), Co-op Option | Ottawa, ON | 3.9 GPA **Sep 2022 – Apr 2027**
- Concentration in Artificial Intelligence and Machine Learning with **$21,000** scholarship for maintaining above **95% average**.
- **Relevant Coursework**: QNX RTOS Training, Algorithm Design & Analysis, OS, DBMS, Web Development, AI, ML

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript/TypeScript, Bash, Linux
Backend & Full Stack: UNIX, Fast API, Flask, Node.js, React.js, HTML/CSS, SQL, PostgreSQL, MySQL, MongoDB
Data & Analytics: ETL Pipelines, Data Processing, Workflows, Reproducible Pipelines, REST APIs, Telemetry Analytics
Test & DevOps: GDB, Visual Studio, Jenkins, Bitbucket, JIRA & Confluence, Postman, CI/CD, Git, Power BI

EXPERIENCE

March Networks - DevOps Software Designing Specialist Co-op | Ottawa, ON **Dec 2024 - Sep 2025**
- **Developed** object-oriented **Python** tools using **REST APIs** to analyze and manage release workflows, improving traceability and reducing manual release errors by **30%** across Linux-based software and firmware systems.
- **Engineered scalable Python analytical workflows** to process runtime telemetry, parse system logs, and automate reproducible validation pipelines integrated within **Jenkins CI/CD environments.**
- **Designed** reusable libraries utilizisng **inheritance and abstraction** for modular data collection, system validation, and diagnostic analysis, significantly bolstering the **reliability** testing and **observability** of complex distributed software systems.
- **Architected** reproducible **Python-based ETL pipelines** integrated with **Jenkins** and **AWS** services to automate telemetry extraction, transformation, and reporting workflows, reducing manual **data processing overhead.**

March Networks - QA Test Engineer Co-op | Ottawa, ON **Sep 2024 - Dec 2024**
- **Investigated** complex system-level failures across **Linux**-based embedded platforms, identifying **35+** critical defects in **C/C++** firmware and camera processing pipelines and **Python test scripts** to verify resolutions and document results.
- **Performed** root-cause analysis using system logs and runtime traces to identify failure conditions, validating engineering fixes by integrating custom scripts with Cypress automation methodologies to increase regression test coverage by **25%.**
- **Recreated** production issues in controlled environments to isolate behaviors, having a **high reproducibility rate** for high-priority bugs, accelerating developer defect resolution via **streamlined diagnostics** and root-cause analysis.
- **Orchestrated** cross-functional testing strategies to validate hardware-software integration, leveraging scripts to audit system health and enhancing final product reliability by **20%** across **high-priority embedded camera firmware releases**.

CDAC- Front-End Web Applications & Interface Development | India **Jan 2023 - May 2023**
- **Designed** and developed responsive web interfaces using HTML, CSS, and JavaScript to build user-friendly dashboards and web applications for enterprise and internal digital platforms supporting **modern browser compatibility.**
- Implemented dynamic UI components and integrated REST APIs using JavaScript and React, improving application usability, navigation flow, and overall front-end performance across multiple devices and screen sizes.
- **Collaborated** with backend developers and UI/UX designers in Agile workflows, used Git for version control, performed debugging and testing and ensured reliable deployment of **stable and maintainable web applications.**

PROJECTS

Smart Mirror AI - Embedded Linux & Intelligent Edge System (C, C++, Python, Linux, AI/ML, Automation)
- **Architected** a modular ecosystem integrating **C/C++** kernels with **Python** microservices to deliver AI-driven fashion analytics and comparative recommendations via an automated chatbot interface optimized for edge processing.
- **Implemented** Linux-based modules to manage **real-time** inter-process communication, strict timing constraints, and deterministic system **behavior** across distributed embedded **software components.**
- **Developed** reproducible Python-based analytical pipelines integrating machine learning heuristics, telemetry processing, and automated performance profiling workflows for **AI-driven decision systems**.
- **Built** automated **test validation suites** to audit system outputs and profile performance metrics, accelerating the resolution of complex integration bottlenecks across distributed low-level software components.

AVAILABILITY : Full-time starting Fall 2026